Lecture 1

CHAPTER 1 FUNCTIONS:

1.1 PREVIEW OF CALCULUS

1.2 FUNCTIONS AND THEIR GRAPHS

1.3 COMBINING FUNCTIONS

1.4 TRIGONOMETRIC FUNCTIONS

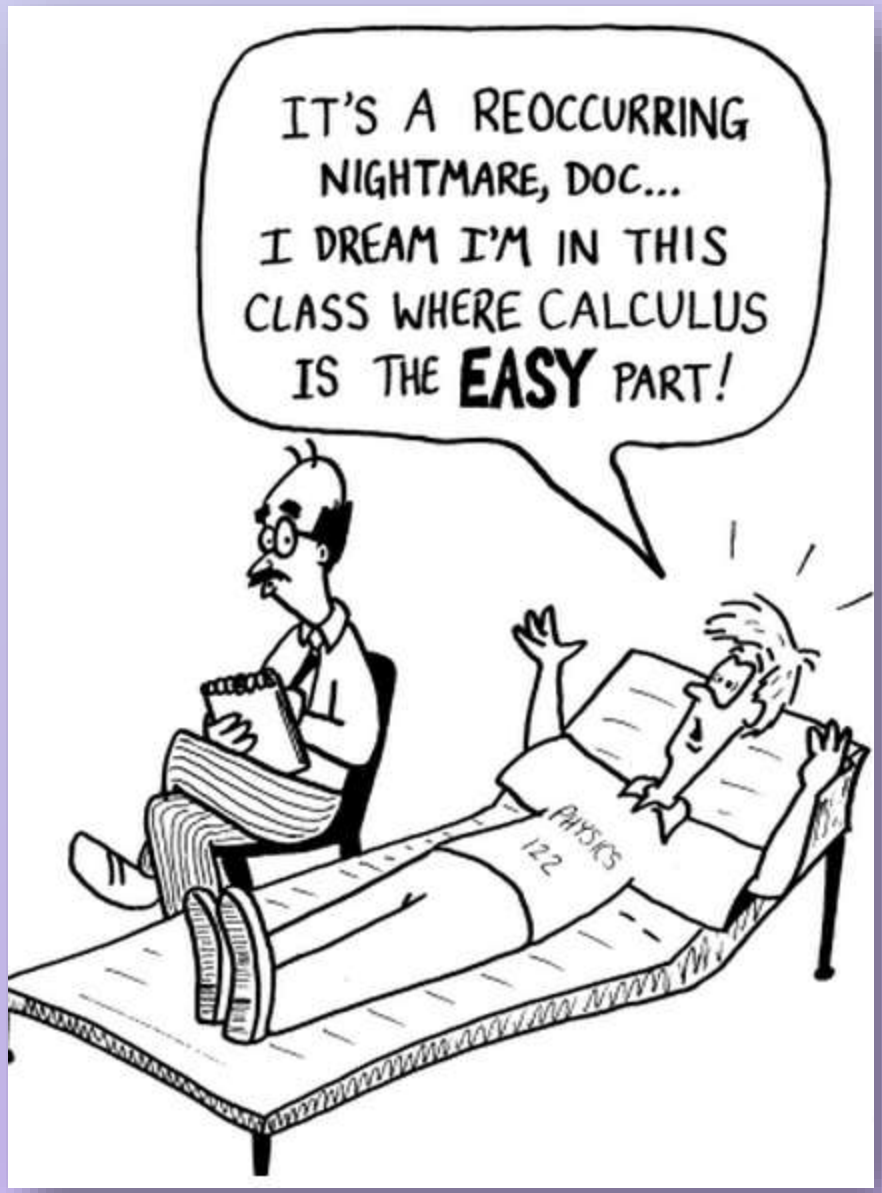
1.5 EXPONENTIAL FUNCTIONS

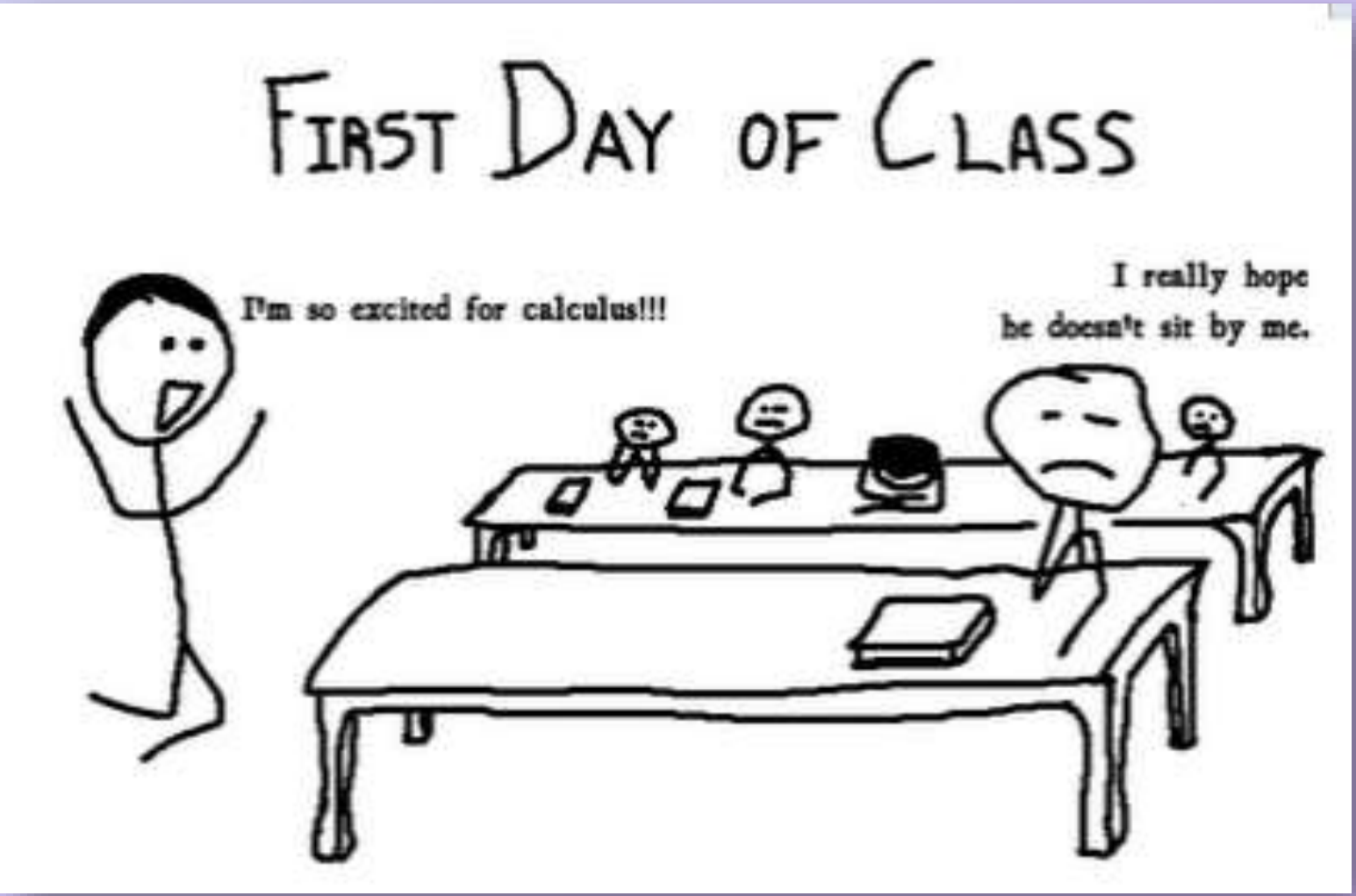
1.1 A Preview of Calculus

- What Do You Think?

It's the Greatest!

- Consider that all these things emerged because of technological advances
- Those advances relied on CALCULUS !
- Calculus has made it possible to:
  - Build giant bridges
  - Travel to the moon
  - Predict patterns of population change


IT'S A REOCCURRING NIGHTMARE, DOC... I DREAM I'M IN THIS CLASS WHERE CALCULUS IS THE EASY PART!
PHYSICS 122


FIRST DAY OF CLASS
I'm so excited for calculus!!!
I really hope
he doesn't sit by me.

The Genius of Calculus is Simple

- It relies on only two ideas
  - The **Derivative**
  - The **Integral**
- Both come from a common sense analysis of motion
  - Motion is change in position over time
  - All you have to do is drop your pencil to see it happen

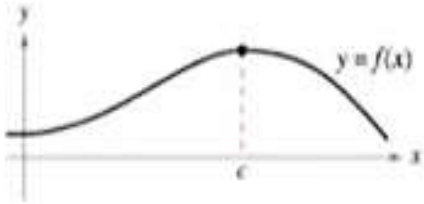
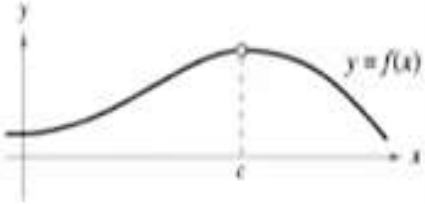
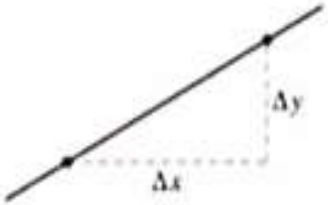
What Is Calculus

- It is the <u>mathematics of change</u>
- It is the mathematics of
  - tangent lines
  - slopes
  - areas
  - volumes
- It enables us to model real life situations
- It is dynamic
  - In contrast to algebra/precalculus which is static

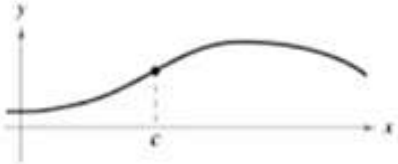
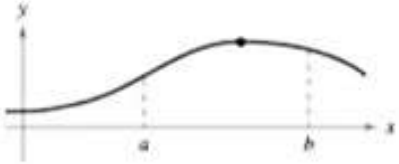
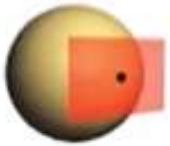
What Is Calculus

- One answer is to say it is a "limit machine"
- Involves three stages
  1. Precalculus/algebra mathematics process
     - Building blocks to produce calculus techniques
  2. Limit process
     - The stepping stone to calculus
  3. Calculus
     - Derivatives, integrals

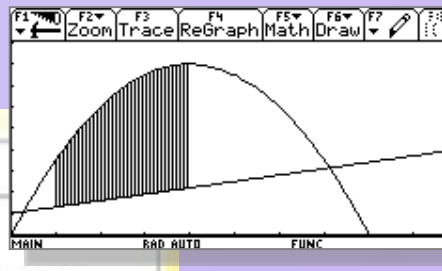
Contrasting Algebra & Calculus

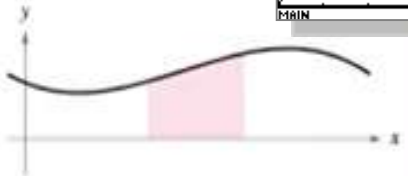
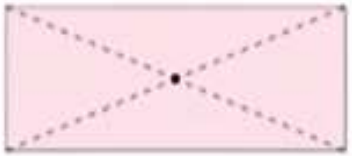
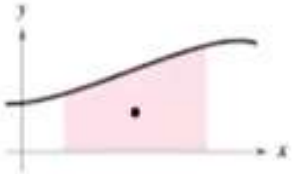
Without Calculus		With Differential Calculus	
Value of $f(x)$ when $x = c$		Limit of $f(x)$ as x approaches c	
Slope of a line		Slope of a curve	
Secant line to a curve		Tangent line to a curve	

Contrasting Algebra & Calculus

Without Calculus	With Differential Calculus
Average rate of change between $t = a$ and $t = b$	Instantaneous rate of change at $t = c$
Curvature of a circle	Curvature of a curve
Height of a curve when $x = c$ 	Maximum height of a curve on an interval 
Tangent plane to a sphere 	Tangent plane to a surface
Direction of motion along a line	Direction of motion along a curve

Contrasting Algebra & Calculus

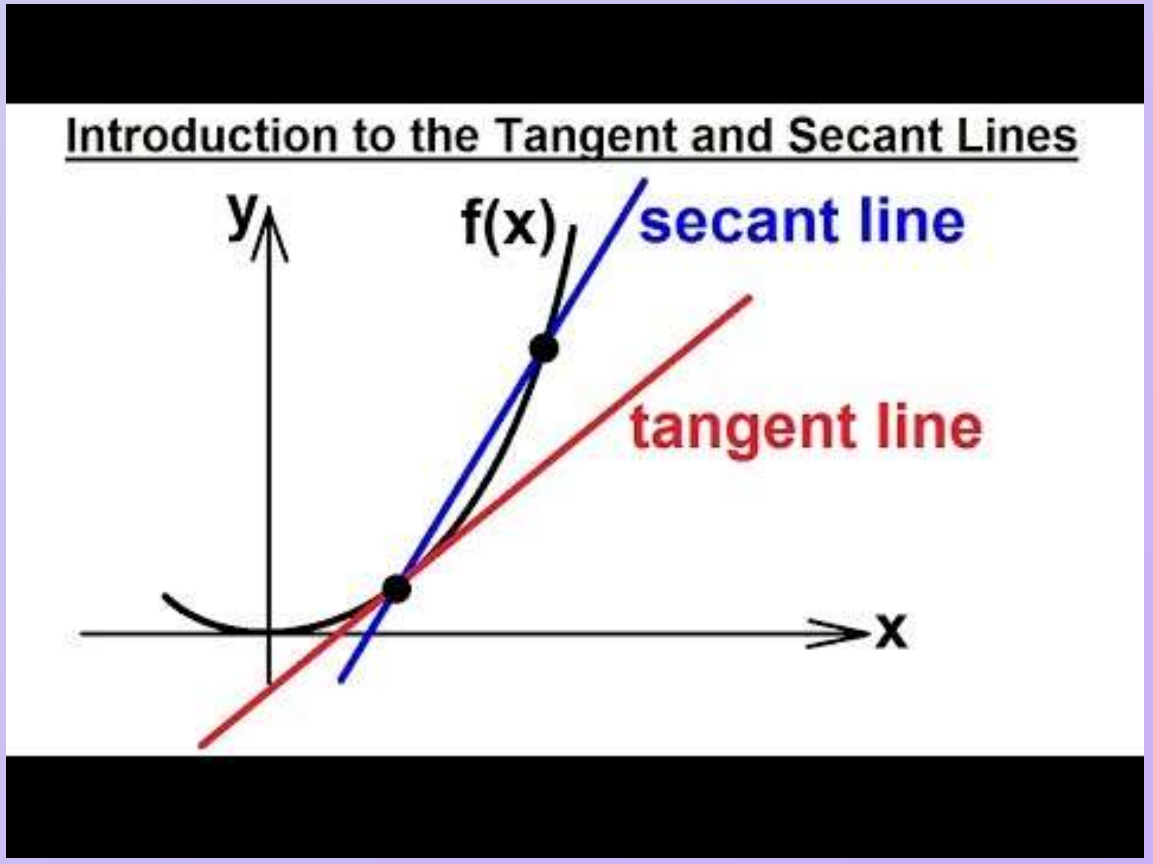


Without Calculus		With Integral Calculus	
Area of a rectangle		Area under a curve **And, area between two curves**	
Work done by a constant force		Work done by a variable force	
Center of a rectangle		Centroid of a region	
Length of a line segment		Length of an arc	

Contrasting Algebra & Calculus

Surface area of a cylinder		Surface area of a solid of revolution	
Mass of a solid of constant density		Mass of a solid of variable density	
Volume of a rectangular solid		Volume of a region under a surface	
Sum of a finite number of terms	$a_1 + a_2 + \cdots + a_n = S$	Sum of an infinite number of terms	$a_1 + a_2 + a_3 + \cdots = S$

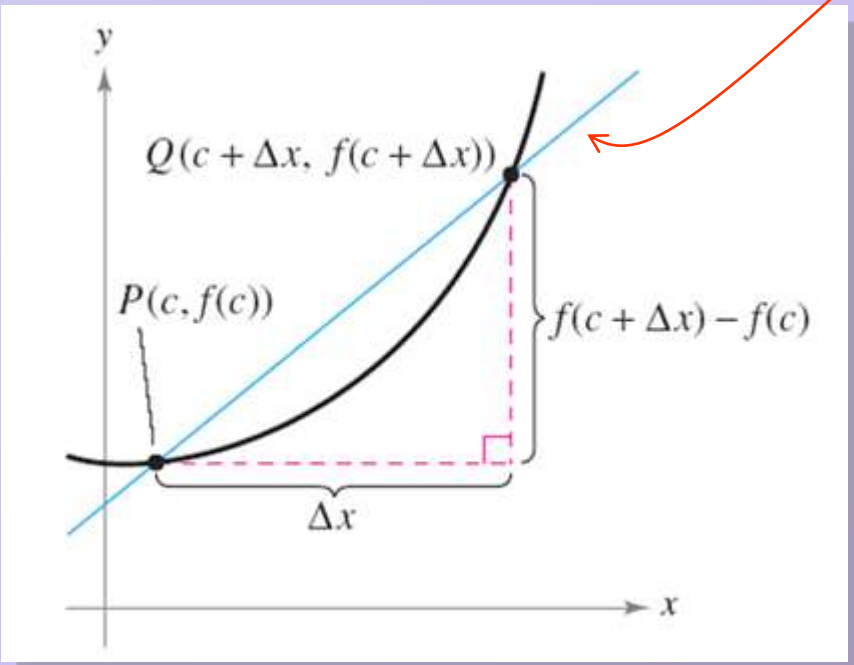
Introduction to Tangent & Secant Lines



Secant line : a line **between** 2 points on a shape
Tangent line: a line **touching** only 1 point on a shape

Classic Problem 1: Tangent Line Problem

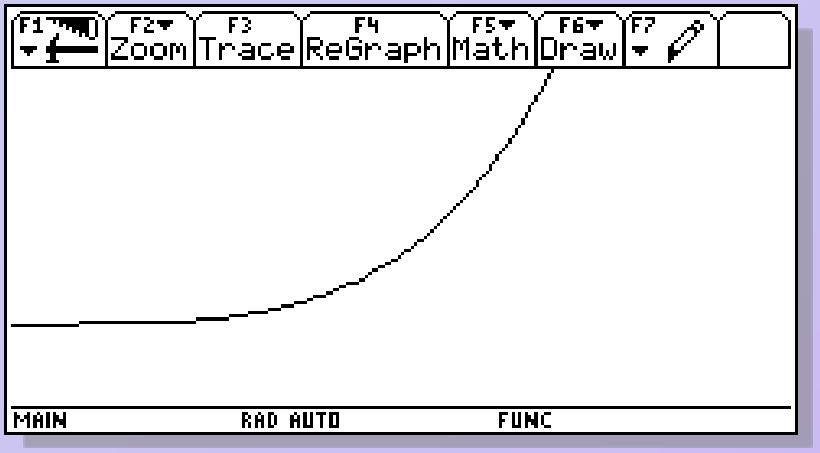
- Approximate slope of tangent to a line
  - Start with slope of secant line



$$m_{\text{secant}} = \frac{f(c+\Delta x) - f(c)}{\Delta x}$$

Tangent Line Problem

- Now allow the Δx to get smaller (when Δx get closer to c)



$$m_{\text{tangent}} = \lim_{\Delta x \to 0} \left(\frac{f(c + \Delta x) - f(c)}{\Delta x} \right)$$

Tangent Line Problem

- Definition of Slope of the Tangent Line

If *f* is defined on an open interval containing x, and if the limit

$$\lim_{\Delta x \to 0}\left(\frac{f(x+\Delta x)-f(x)}{\Delta x}\right) = m \text{ exists,}$$

then the line passing through (x, $f(\Delta x)$) with slope m is the tangent line to the graph of f at the point (x, $f(\Delta x)$). This slope of the tangent line to the graph of f at the point (x, $f(\Delta x)$) is also called the slope of the graph of f at x = Δx.

Tangent Line Problem

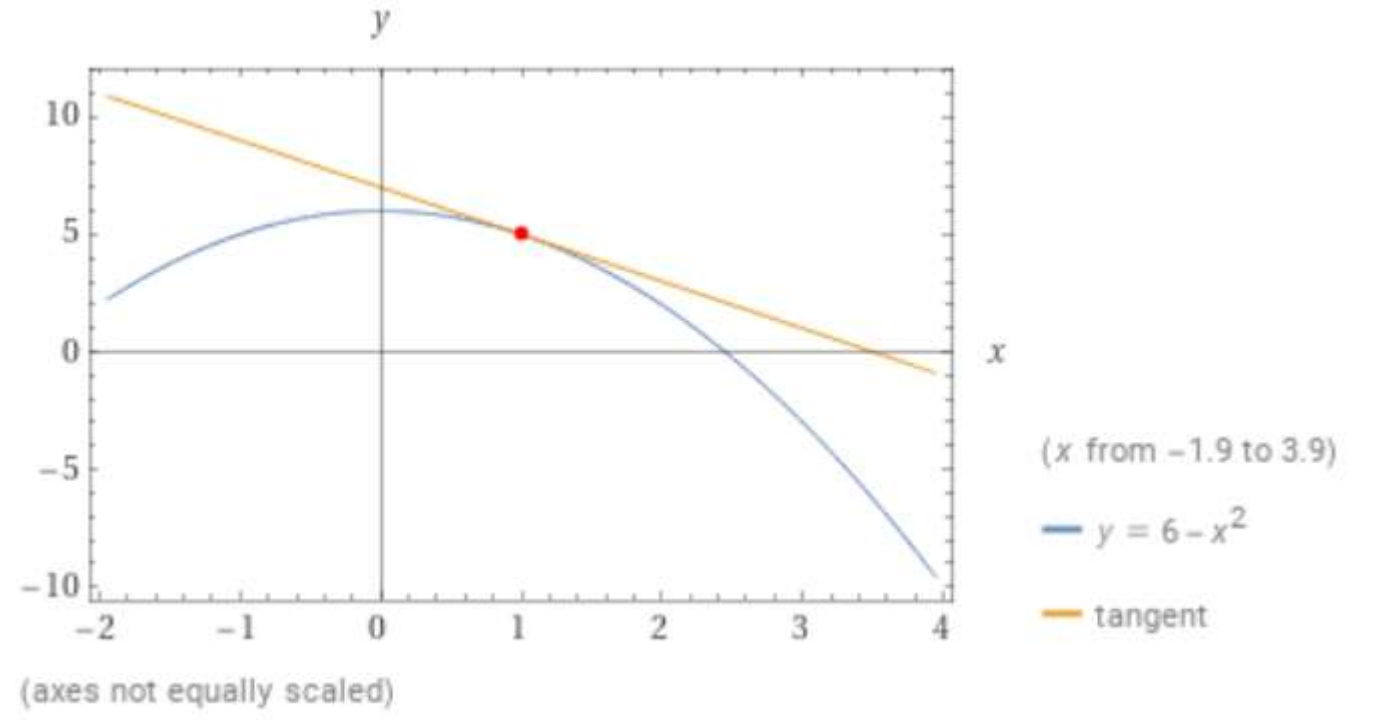
- Example: (a) Find the slope of the tangent line to the graph of the function at the given point. (b) Find an equation of the tangent line to the graph of the function. (c) Use a graphing utility to graph the function and its tangent line at the point.

1. $g(x) = 6 - x^2$; (1, 5)

(a) $-2x$

(b) $y = -2x+7$

(c)


y
x
10
5
0
−5
−10
−2
−1
0
1
2
3
4
(x from −1.9 to 3.9)
y = 6 − x^2
tangent
(axes not equally scaled)

Tangent Line Problem

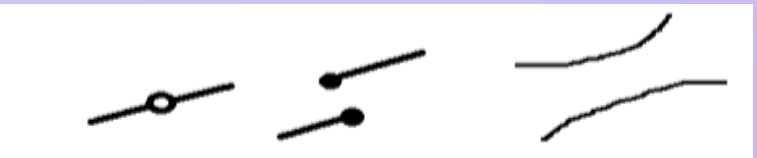
- Definition of the Derivative of a Function

The derivative of f at x is $f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x)-f(x)}{\Delta x}$

provided the limit exists. For all x for which this limit exists, $f'(x)$ is a function of x.

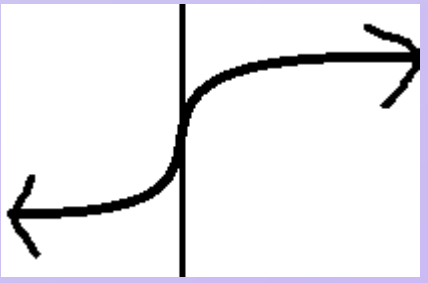
Location where tangent lines DO NOT Exist

1. Discontinuities: holes/jumps/asymptotes



2. Sharp turns

3. Vertical tangent lines



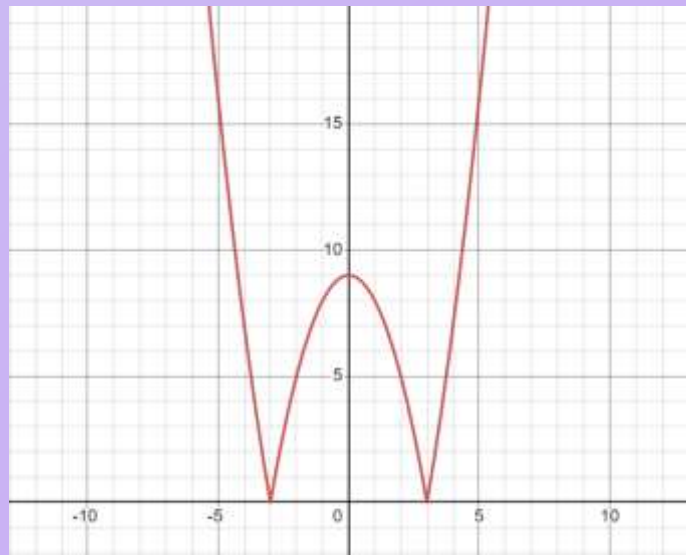
$x = 2$

Theorem: Differentiability Implies Continuity

- If f is differentiable at $x = c$, then f is continuous at $x = c$.

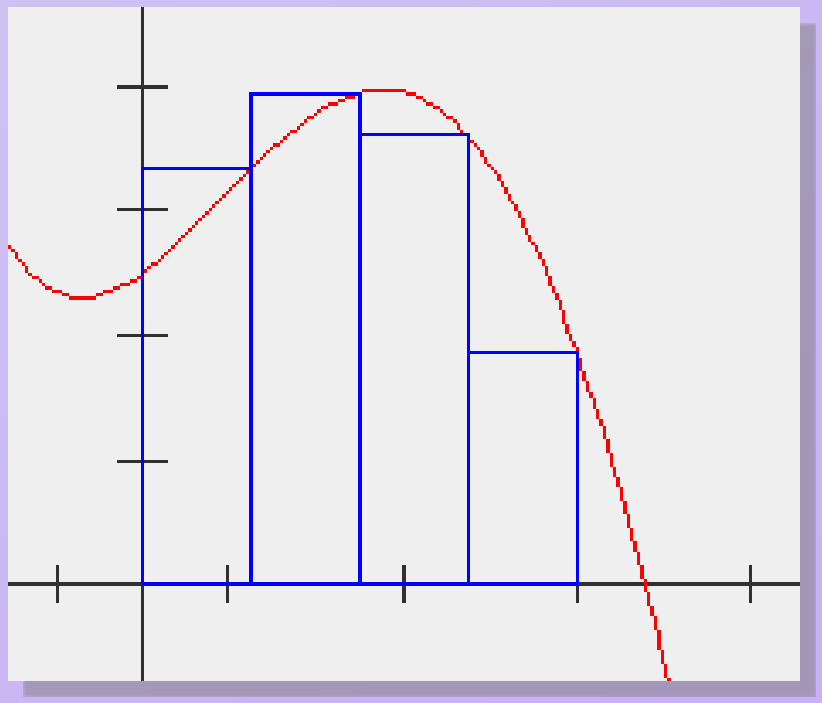
- Examples: Describe the x-values at which f is differentiable.

4. $f(x) = |x^2 - 9|$



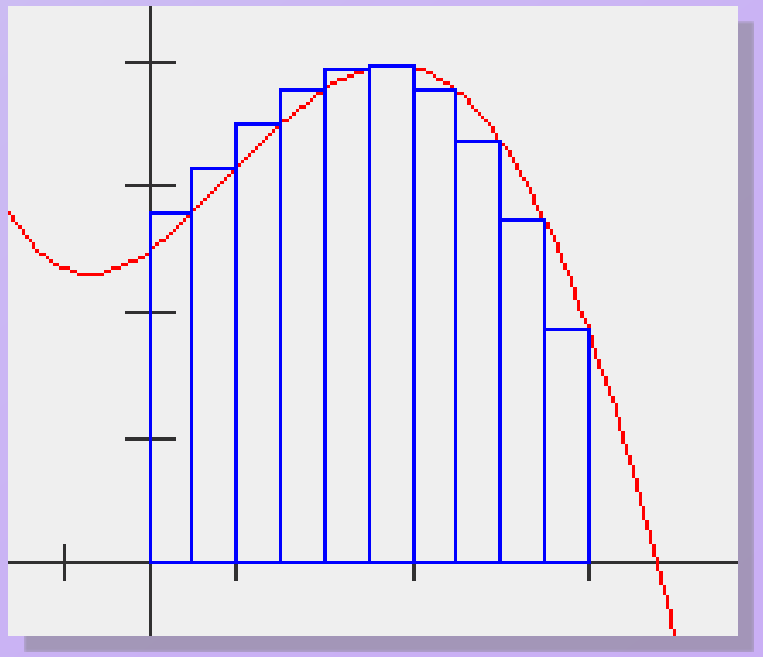
Classic Problem 2:The Area Problem

- We seek the area under a curve, the graph f(x).
- We approximate that area with a number of rectangles.
- Sum = 31.9
- Actual = 33.33



The Area Problem

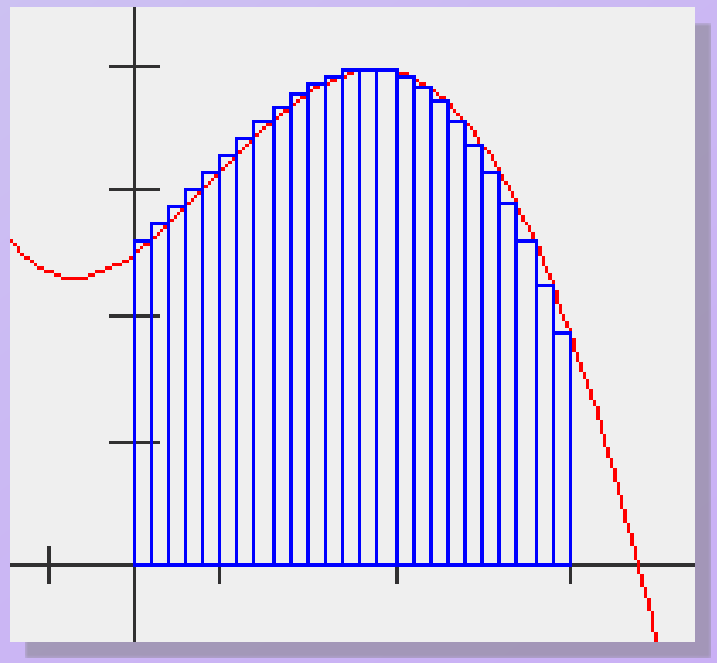
- The approximation is improved/becomes better and better by increasing the number of rectangles (the amount of area missed by the rectangles is decreased.)
- Number of rectangles = 10
- Sum = 32.92
- Actual = 33.33



The Area Problem

- The approximation is improved by increasing the number of rectangles.

- Number of rectangles = 25
- Sum = 33.19
- Actual = 33.33

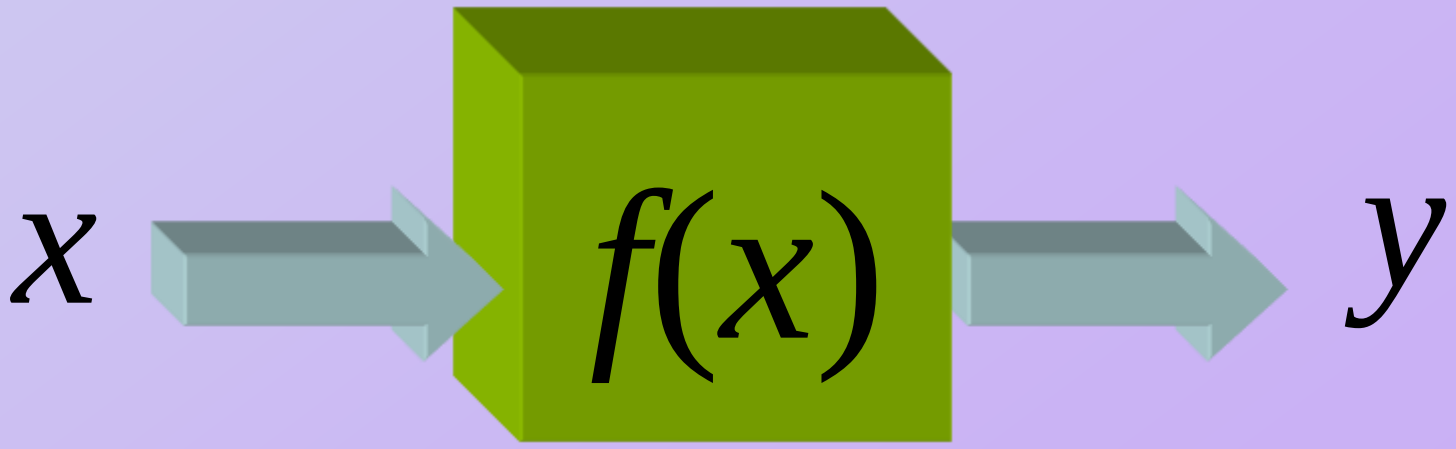


1.2 Functions and their graphs

- Objectives:
  - To expose to various types of functions
  - To identify domain and range of a function
  - To sketch graphs of functions

Functions

A **function** is a relation in which each element of the domain is paired with *exactly one* element of the range. Another way of saying it is that there is one and only one output (y) with each input (x).



Function Notation

$$y = f(x)$$

Input

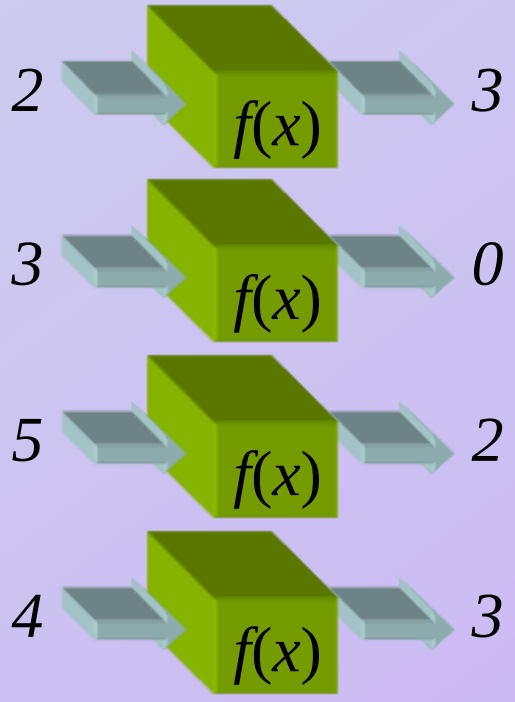
Output

Name of Function

x is the independent variable
y is the dependent variable

Determine whether each relation is a function.

1. {(2, 3), (3, 0), (5, 2), (4, 3)}

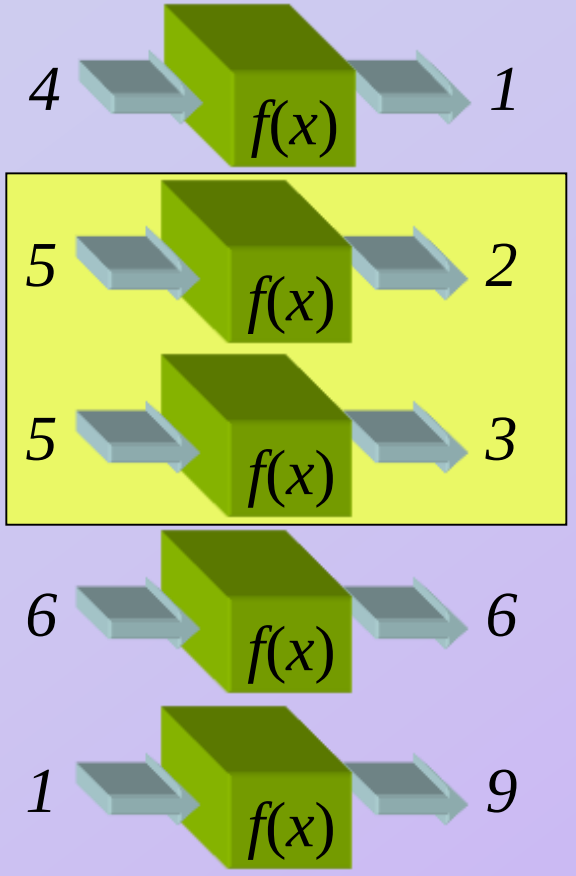


$(x\ values)$

YES, every <u>domain</u> is different!

Determine whether the relation is a function.

2. {(4, 1), (5, 2), (5, 3), (6, 6), (1, 9)}



NO,

5 is paired with 2 numbers!

Is this relation a function?

{(1,3), (2,3), (3,3)}

1. Yes ✓
2. No

Answer Now

Tell whether the equations represent *y* as a function of *x*.

a. $x^2 + y = 1$

Solve for *y*.

$y = 1 - x^2$

For every number we plug in for *x*, do we get more than one *y* out?

No, so this equation is a function.

b. $-x + y^2 = 1$

Solve for *y*.

$y^2 = x + 1$

$$y = \pm\sqrt{1+x}$$

Here we have 2 *y*'s for each *x* that we plug in. Therefore, this equation is not a function.

Domain and Range of Function

- **Domain** – set of ***x*-values**.
- For a function $f : X \rightarrow Y$ the **domain** of f is the set X.
  - *e.g.* $y = \sqrt{x}$ has domain; all real $x \geq 0$.
- **Range** – set of ***y*-values**.
- For a function $f : X \rightarrow Y$ the **range** of f is the set of y-values such that $y = f(x)$ for some x in X.
  - e.g. $y = \sqrt{x}$ has range; all real $y \geq 0$.

Find the domain of each function.

a. f: {(-3,0), (-1,4), (0,2), (2,2), (4,-1)}

Domain = { -3, -1, 0, 2, 4}

b. $$g(x) = \frac{1}{x+5}$$

D: $\Re, x \neq -5$

c. $$f(x) = \sqrt{4 - x^2}$$

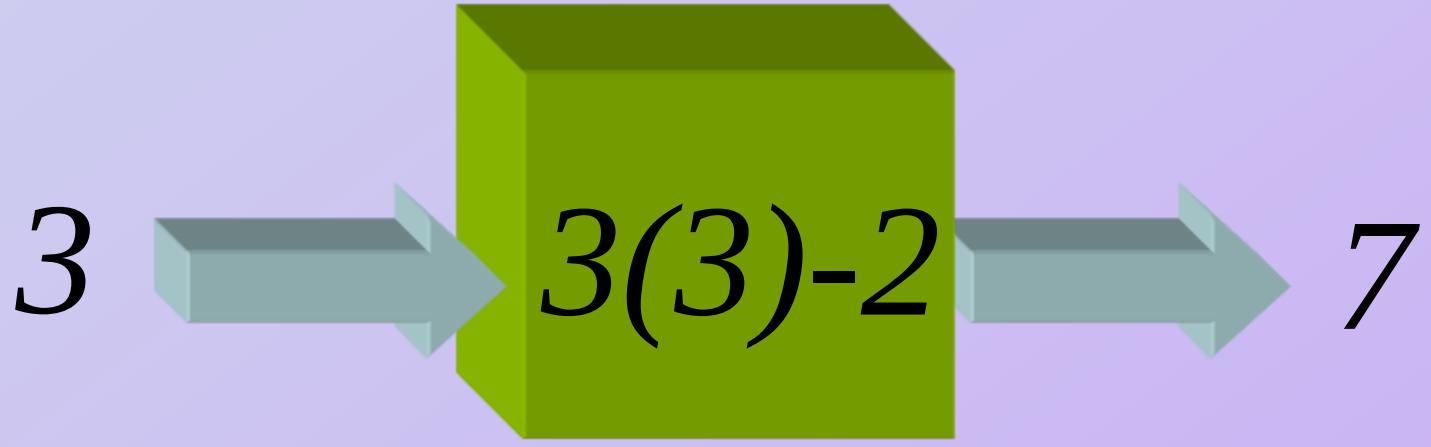
Set 4 – x^2 greater than or = to 0, then factor

$$(2-x)(2+x) \geq 0$$

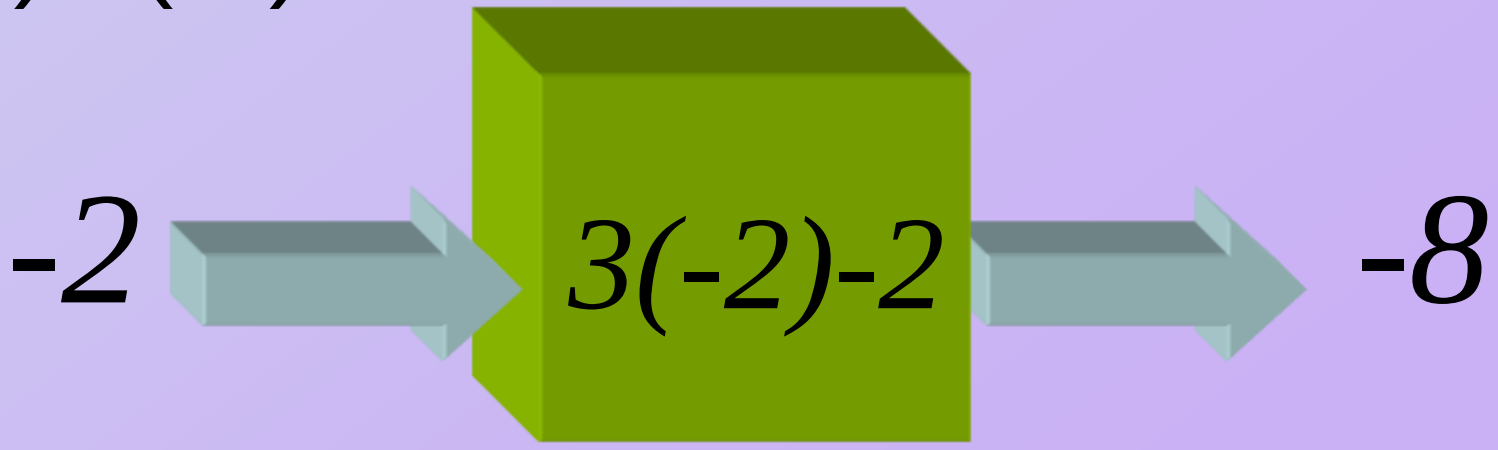
D: [-2, 2]

Given f(x) = 3x - 2, find:

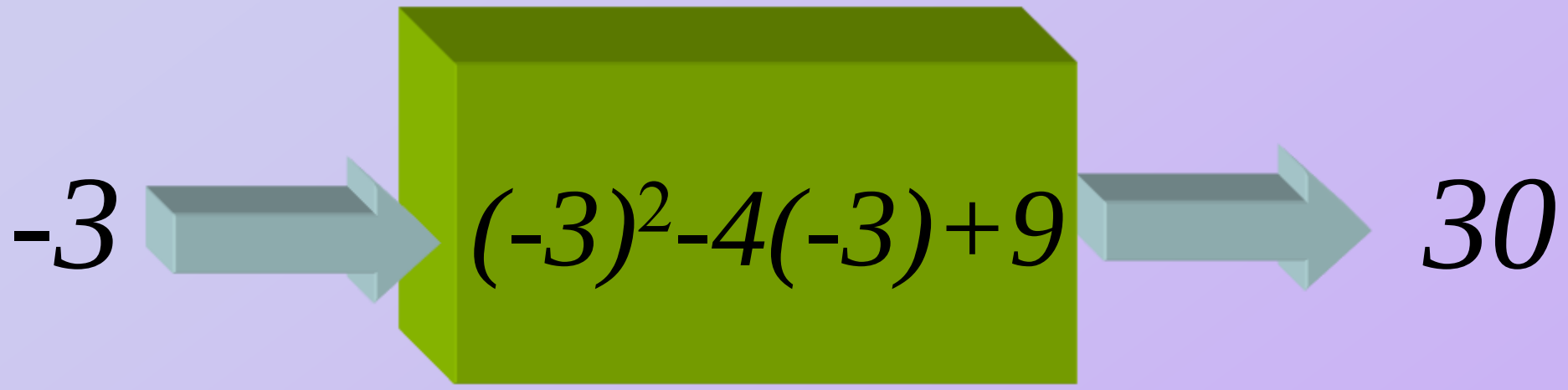
1) $f(3) = 7$



2) $f(-2) = -8$



Given $h(z) = z^2 - 4z + 9$, find $h(-3)$



$$9 + 12 + 9$$

$$h(-3) = 30$$

Given $g(x) = x^2 - 2$, find $g(4)$

1. 2
2. 6
3. 14
4. 18

Answer Now

Given $f(x) = 2x + 1$, find $-4[f(3) - f(1)]$

1. -40
2. -16
3. -8
4. 4

Answer Now

Ex. $g(x) = -x^2 + 4x + 1$

Find: a. $g(2)$

b. $g(t)$

c. $g(x+2)$

d. $g(x + h)$

Ex.

$$f(x) = \begin{cases} x^2 + 1, x < 0 \\ x - 1, x \geq 0 \end{cases}$$

Evaluate at x = -1, 0, 1

Ans. 2, -1, 0

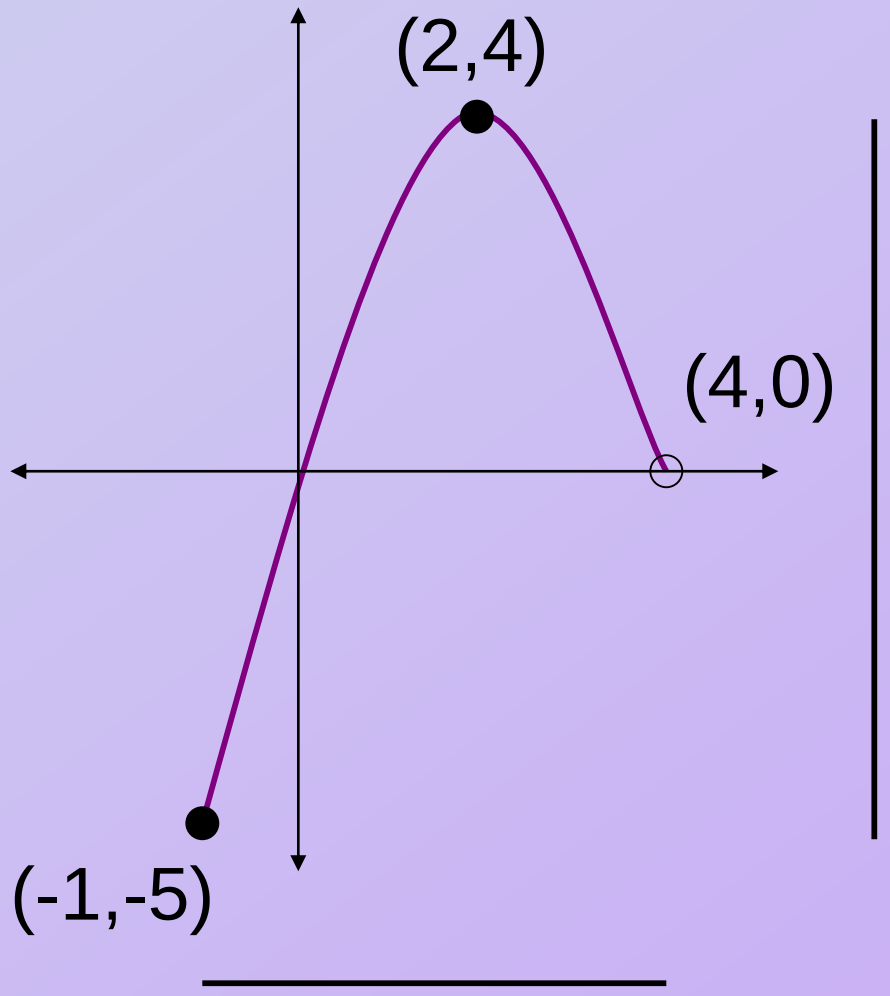
Find:

a. the domain
   [-1,4)

b. the range
   [-5,4]

c. f(-1) = -5

d. f(2) = 4



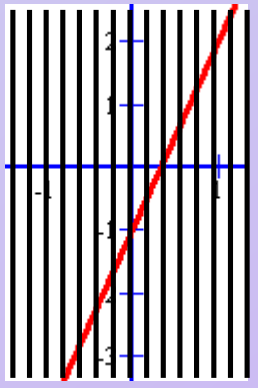
- Verify the natural domains and associated ranges of these functions

Function	Domain (x)	Range (y)
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = \frac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$y = \sqrt{4 - x}$	$(-\infty, 4]$	$[0, \infty)$
$y = \sqrt{1 - x^2}$	$[-1, 1]$	$[0, 1]$

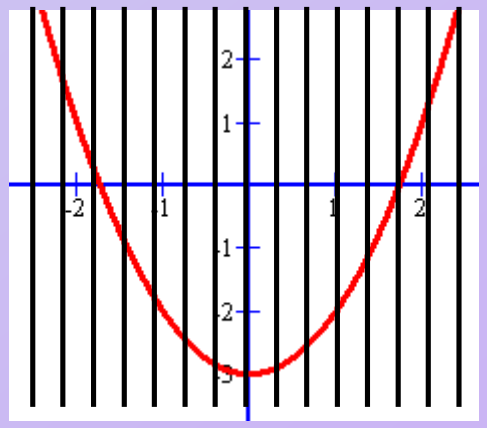
Vertical Line Test (pencil test)

If any vertical line passes through more than one point of the graph, then that relation is not a function.

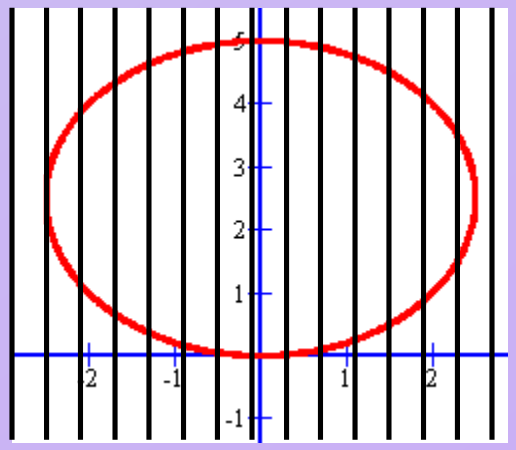
Are these functions?



FUNCTION!

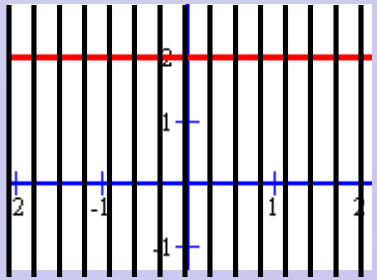


FUNCTION!

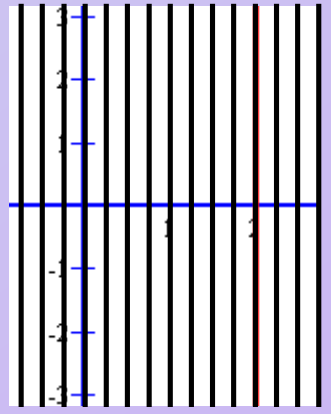


NOPE!

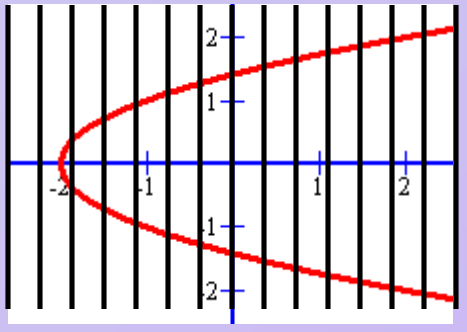
Vertical Line Test



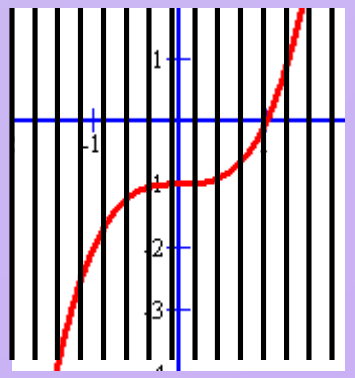
FUNCTION!



NO!

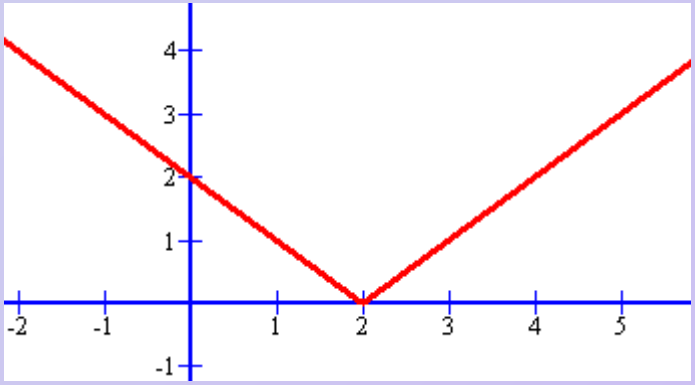


NO WAY!



FUNCTION!

Is this a graph of a function?

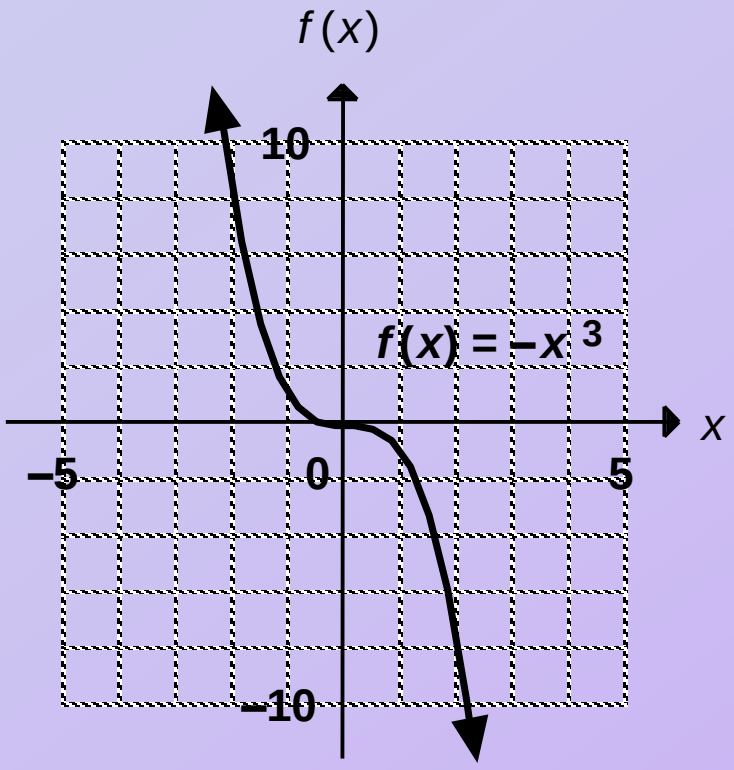


✔ 1. Yes

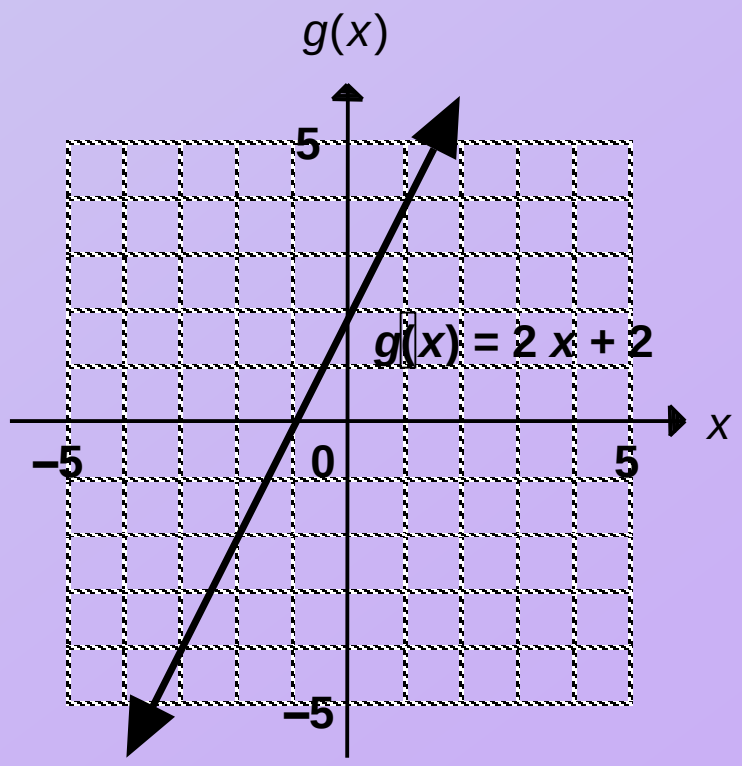
2. No

Answer Now

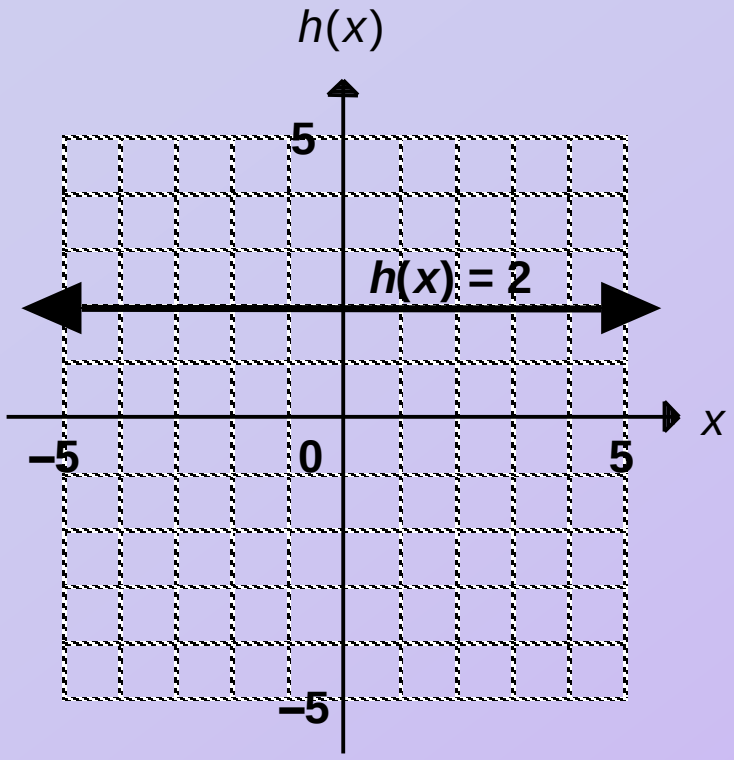
Increasing, Decreasing, and Constant Functions



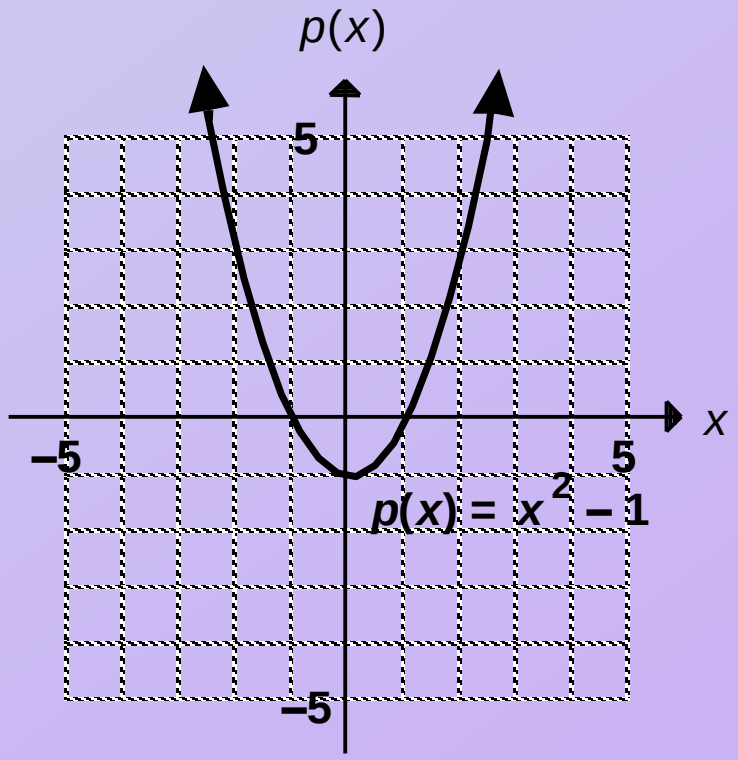
(a) Decreasing on $(-\infty, \infty)$



(b) Increasing on $(-\infty, \infty)$



(c) Constant on (–∞, ∞)

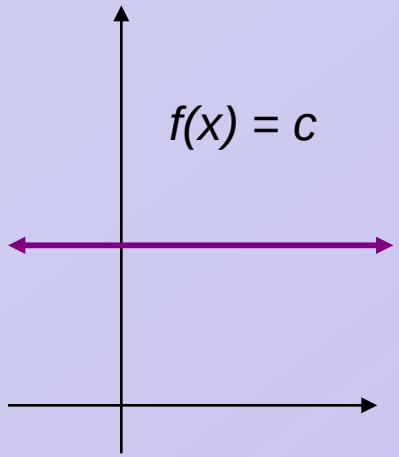


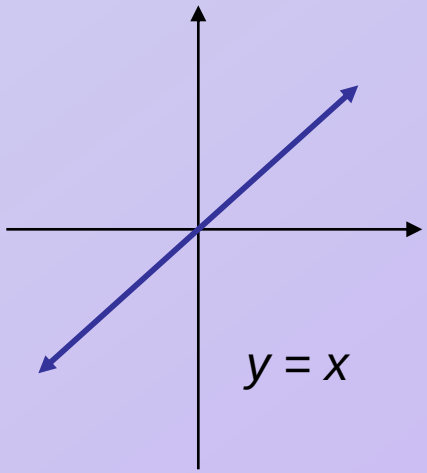
(d) Decreasing on (–∞, *0*]
Increasing on [0, ∞)

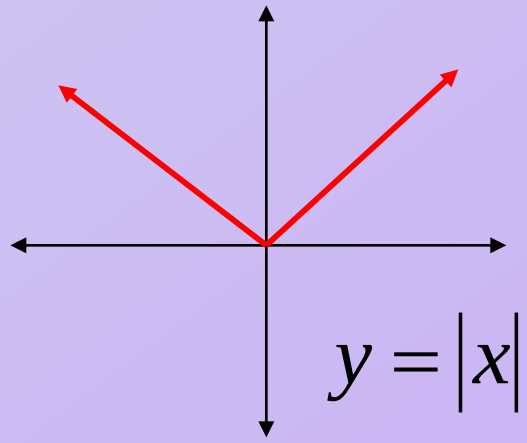
Types of Function

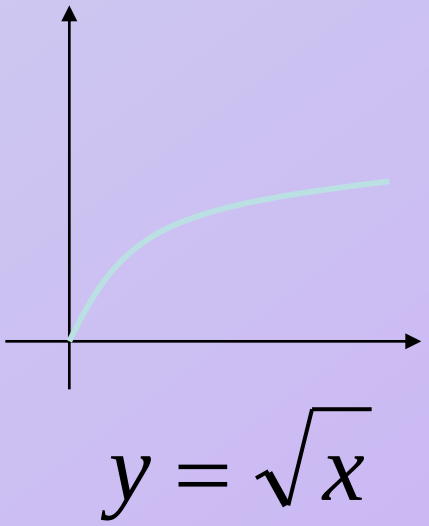
- Constant function
- Power function
- Linear function
- Quadratic function
- Cubic function
- Polynomial function
- Rational function
- Piecewise-defined function
- Trigonometric function
- Exponential function
- Logarithmic function

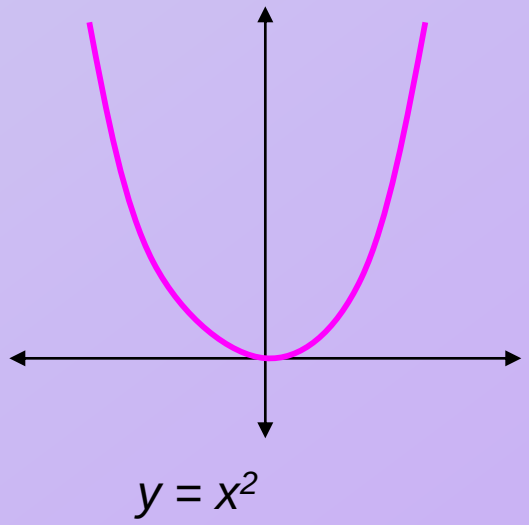
Graphs of Common Functions

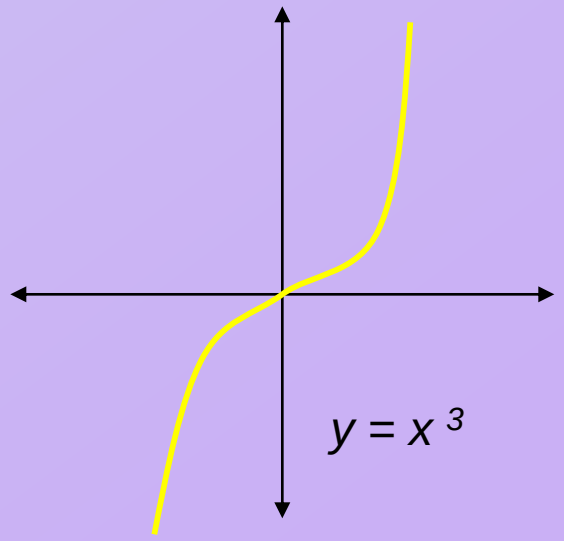












Tests for Even and Odd Functions

A function of $y = f(x)$ is <u>even</u> if, for each x in the domain of f,

$$f(-x) = f(x)$$

An even function is symmetric about the y-axis.

A function of $y = f(x)$ is <u>odd</u> if, for each x in the domain of f,

$$f(-x) = -f(x)$$

An odd function is symmetric about the origin.

Ex. $g(x) = x^3 - x$

$$g(-x) = (-x)^3 – (-x) = -x^3 + x = -(x^3 – x)$$

Therefore, $g(x)$ is odd because $g(-x) = -g(x)$

Ex. $h(x) = x^2 + 1$

$$h(-x) = (-x)^2 + 1 = x^2 + 1$$

$h(x)$ is even because $h(-x) = h(x)$

1.3 Combining Functions

- Objective:
  - To be able to combine or transform functions to form new functions

1.3 Combining Functions

- Functions can be added, subtracted, multiplied and divided (except when the denominator is zero) to produce new functions.
- If f and g are functions and x belongs to the domains of f and g, we define:

$$(f+g)(x)= f(x)+g(x)$$

$$(f-g)(x)=f(x)-g(x)$$

$$(fg)(x)=f(x)g(x)$$

1.3 Combining Functions

- We can also define at any point in the domains of *f* and *g* at which $g(x) \neq 0$:

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \text{ where } g(x) \neq 0.$$

- Functions can also be multiplied by constants. If *c* is a real number, then the functions *cf* is defined for all *x* in the domain of

$$(cf)(x)=cf(x)$$

Composite Functions

The composition of the functions *f* and *g* is

$$(f \circ g)(x) = f(g(x))$$

"*f* composed by *g* of *x* equals *f* of *g* of *x*"

- The domain of $(f \circ g)$ consists of the numbers *x* in the domain of *g* for which *g(x)* lies in the domain of *f*

Ex. $f(x) = \sqrt{x}$, $g(x) = x - 1$.

Find $(f \circ g)(2)$

$$(f \circ g)(x) = f(g(x)) = \sqrt{x-1},$$

$$(f \circ g)(2) = \sqrt{2-1} = 1.$$

Ex. $f(x) = x + 2$ **and** $g(x) = 4 - x^2$. **Find:**

$$(f \circ g)(x) = f(g(x)) = (4 - x^2) + 2 = -x^2 + 6$$

$$(g \circ f)(x) = g(f(x)) = 4 - (x + 2)^2$$

$$= 4 - (x^2 + 4x + 4) = -x^2 - 4x$$